

TITLE

Estimating Overall and Heterogeneous Effects of Digital Health Messages

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INTRODUCTION AND AIMS OF THE STUDY

This section provides an overview of the context of the research question, including what is known and unknown, and requires a detailed literature search. This should contain the following four elements, and will be used to construct the Introduction and Discussion sections of the manuscript:

1. Define the problem or research question.
2. Describe what is known about the research relative to what you plan to study. For example, what studies have already examined an aspect of this research question, and what remaining questions exist that this research project will address.
3. What are the gaps in knowledge that this study will address (usually 2–3).
4. Describe the hypotheses of the study questions.

DELIVERABLE TYPE AND DEADLINE (OPTIONAL)

_____	Abstract	Deadline Date: _____
_____	Manuscript	Deadline Date: _____
X	Internal Report	Deadline Date: Oct 21, 2026
_____	Pilot data for grant	Deadline Date: _____

_____ Other (please explain): _____

Deadline Date: _____

STUDY COHORT

PROMPT and COMPASS Cohorts Diagram

- | | |
|---|---|
| ☐ RCT PROMPT | ☐ Ro1 PROMPT |
| ☐ COMPASS | ☐ Full Cohort |
| ☒ 2025 IHS Cohort | ☒ 2024 IHS Cohort |

We will prioritize 2025 IHS cohort, possibly extending to the 2024 cohort if time permits.

DATA TYPES (E.G. SURVEY, FITBIT, GENOMIC, SENSORKIT)

Survey (for BL characteristics), Fitbit & Garmin & HealthKit (for steps/sleep/mood outcomes as well as moderators)

INCLUSIONS (E.G. SCREENING CUTOFFS, APP ASSIGNMENT)

N/A

EXCLUSIONS (IF ANY)

We may exclude those who have too much missing data.

VARIABLES

MAIN EXPOSURE

Notification (send vs no send), notification type (sleep vs mood vs steps), notification strategy (therapeutic strategies)

PRIMARY OUTCOME

Your text here

SECONDARY OUTCOME (s)

Your text here

ADDITIONAL COVARIATES (PLEASE SPECIFY IF THESE WILL BE CONSIDERED AS CONFOUNDERS, OR MEDIATING AND MODERATING RELATIONSHIPS WILL ALSO BE EXPLORED)

Your text here

STATISTICAL ANALYSIS

This can be your fully established analysis, or draft prior to potential consult with our analysis team.

Univariate approach:

Multivariable approach:

Our proposed approach is to use DR-WCLS to estimate the global population average effect. Then, the DR-WCLS output will allow us to regress moderators on individual pseudo-outcomes to estimate the individual baseline treatment effects. We will then compute the significance of effects and compare significance of effects across individuals with null distribution derived from permuting the randomization to determine if the effect is indeed significant overall across individuals. Looking at the distribution of effects, we will determine heterogeneity by fitting Gaussian Mixture Models. If the effect across individuals are indeed heterogeneous, we will analyze for what individuals the effect is positive vs negative using baseline characteristics.

LIMITATIONS OF THE STUDY (OPTIONAL)

Preprocessing method for timeseries variable. No clear distinction between users of each device type. Harmonization across device types ad hoc.

Title, description and shell of tables and mock figures. The following are examples for Table 1 and figures.

1. Table 1 – Characteristics

Characteristic	Overall (No. of Patients, %)	Garmin (No. Patients, % Total)	HealthKit (No. Patients, % Total)	Fitbit (No. Patients, % Total)
Race				
White				
Black				
Others				
Age				
...	...	...	...	...